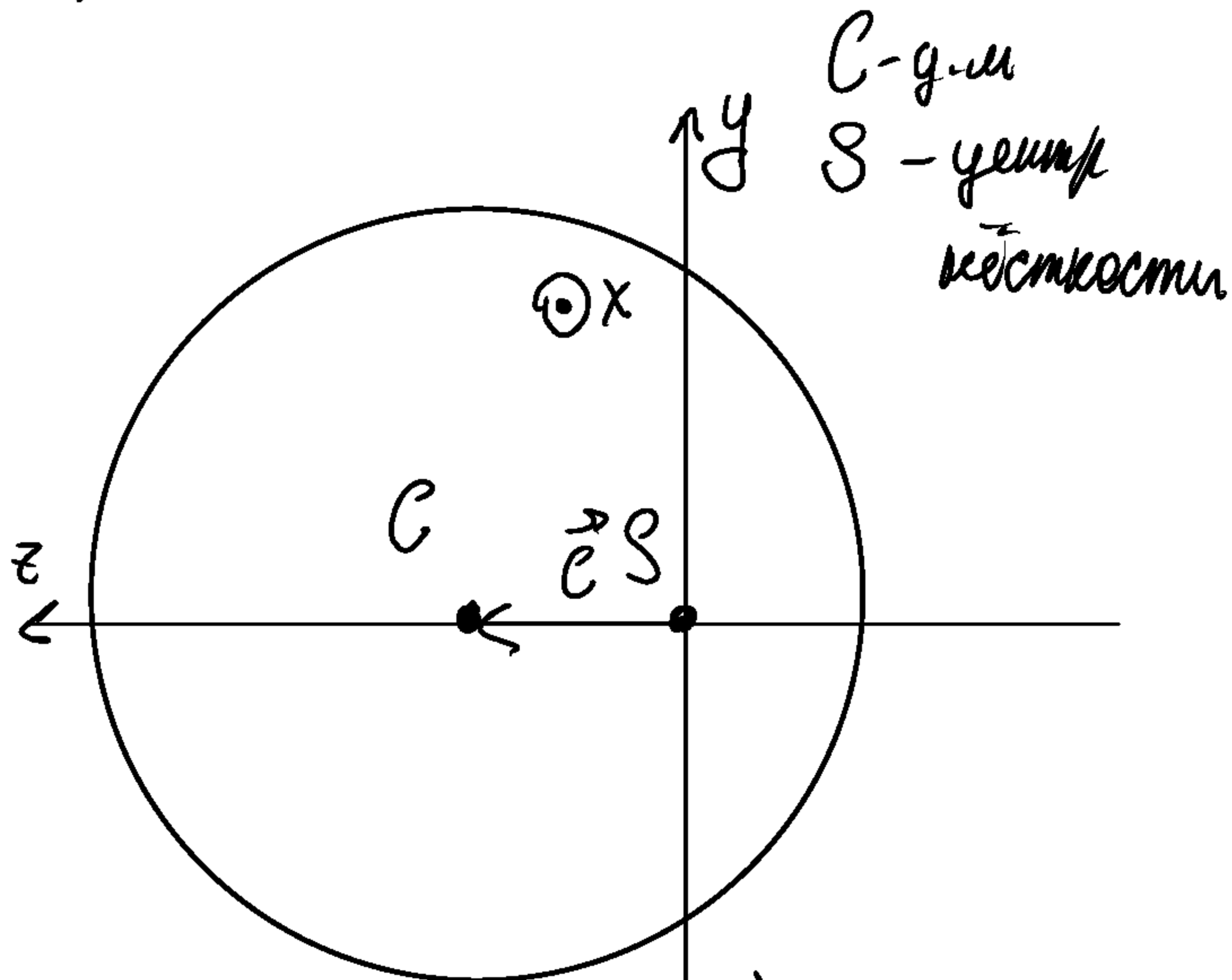


1) Для сегмента с крив. симметрией
центр тяжести и центр масс не
совпадают



$$\vec{r}_e \approx \vec{r}_g + \vec{e} + \varphi[\vec{x}, \vec{e}]$$

$$\vec{r}_8 = (0, 4H, 0)$$

$$[\hat{x}, \vec{e}] = (0, -c_z, c_y)$$

$$\vec{r}_0 = (0, 4 - \rho e_z, \frac{\rho}{\sin \theta})$$

$$y_e = U - \varphi C_z \Rightarrow \dot{y}_e = \dot{U} - C \dot{\varphi}$$

$$dT_{\text{сеч}} = \frac{1}{2} (dm \dot{y}_c^2 + dI_c \dot{\varphi}^2)$$

$$dm = \rho S dx, dI_c = \rho I_p dx$$

$$dT_{\text{сеч}} = \frac{1}{2} dm \dot{u}^2 - dm e \dot{u} \dot{\varphi} + \frac{1}{2} (dI_c + dm c^2) \dot{\varphi}^2$$

$$T_{\text{сеч}} = \frac{1}{2} \int_0^L \left[\rho S \dot{u}^2 - 2 \rho S e \dot{u} \dot{\varphi} + \rho I_p \dot{\varphi}^2 \right] dx$$

2) Поиск ном.-энергии

$$u_x(y, z, t, x)$$

$$u_y(y, z, t, x)$$

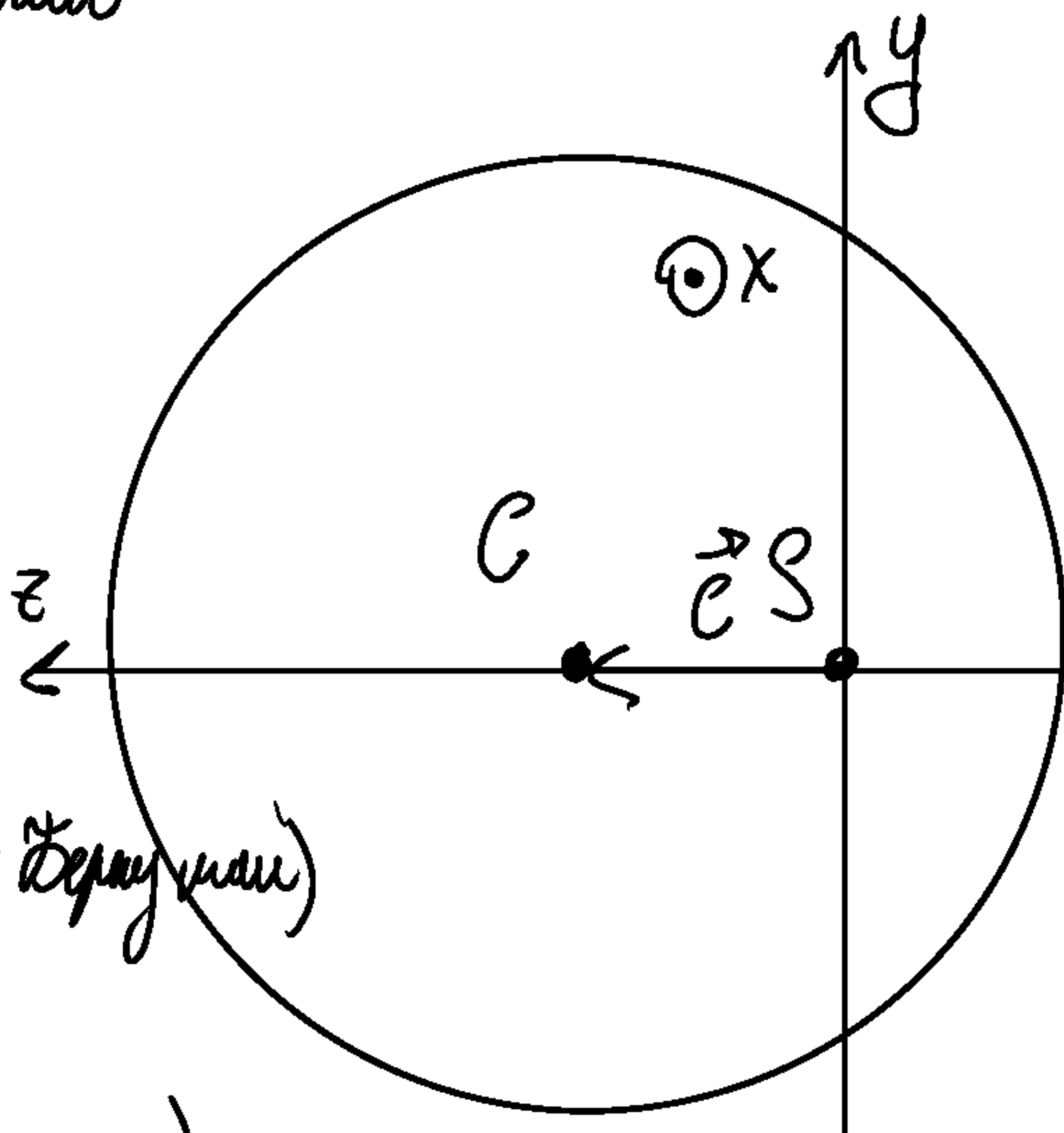
$$u_z(y, z, t, x)$$

$$u_y^{\text{об}}(x, t) = w(x, t) \quad (\text{Эйлер-Бернулли})$$

$$u_x = -y \partial_x w$$

(Перемещение S плоское)

$$u_y = w(x, t) - z \varphi(x, t), u_z = y \varphi(x, t)$$



$$\varepsilon_{xx} = \partial_x u_x = -\nu \partial_x^2 w$$

$$\partial_y u_y = 0 = \partial_z u_z$$

Кручение даёт связь

$$f_{x\theta} = r \frac{\partial \varphi}{\partial x} \quad r - \text{расст. до } OX, \theta - \text{угл. коэф.}$$

$$f_{xy} = -z \frac{\partial \varphi}{\partial x}$$

$$f_{xz} = y \frac{\partial \varphi}{\partial x}$$

$$\varepsilon_{yy} = \varepsilon_{zz} = -\nu \varepsilon_{xx} = -\nu \partial_x^2 w$$

Законы Гука

$$\sigma_{xx} = E \varepsilon_{xx} = -E \nu \partial_x^2 w$$

$$\tau_{xy} = G f_{xy} = -G z \partial_x \varphi$$

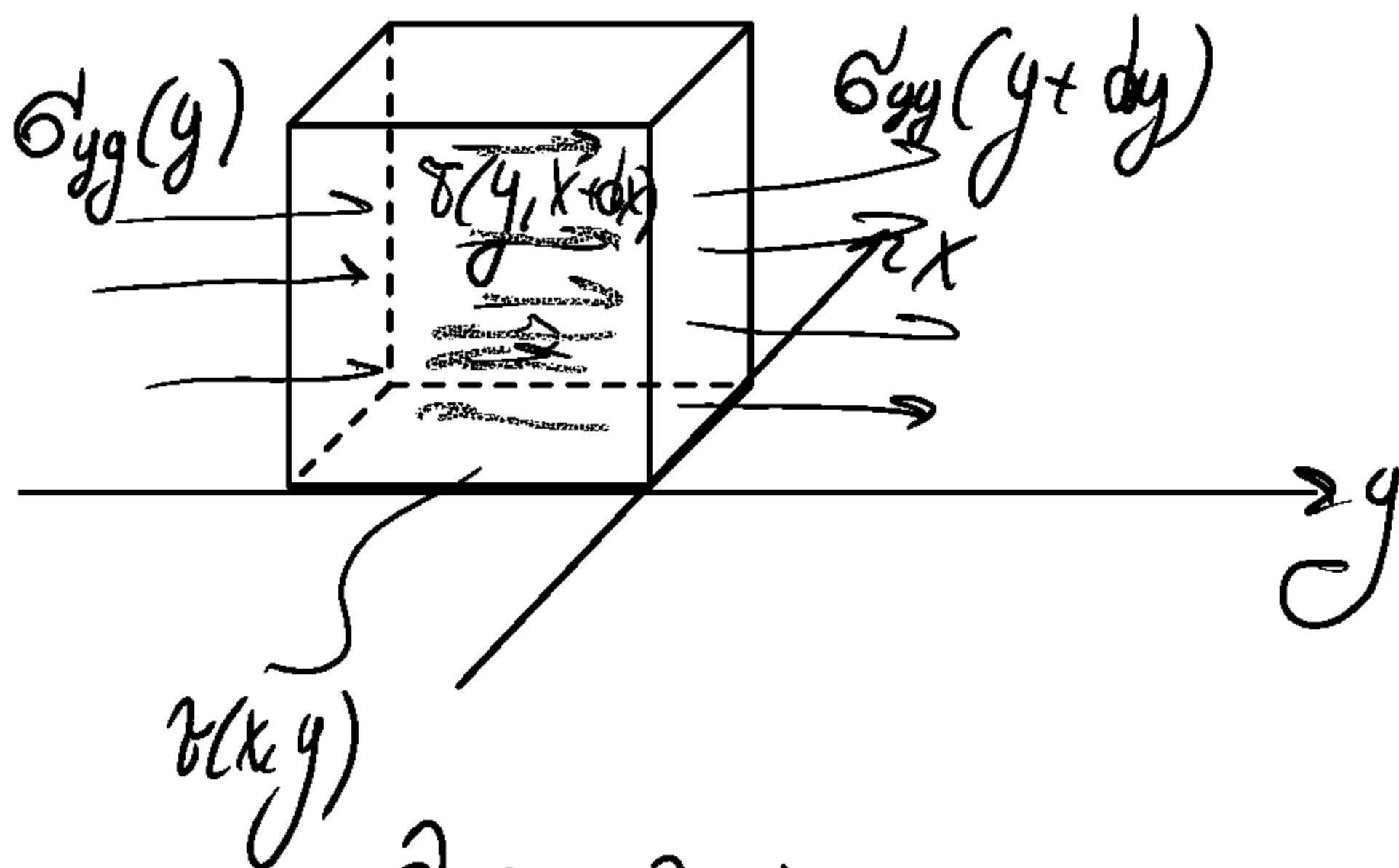
$$\tau_{xz} = G f_{xz} = G y \partial_x \varphi$$

u_z свобод. деф. пов-сти

$$\sigma'_{yy} = \sigma'_{zz} = 0. \text{ Это верно для}$$

кв. энергии, но я считаю это верным и
 дам кривую энергии в силу ее точности

Рассм. элем. энергии



$$\partial_x \delta + \partial_y \sigma_{yy} = 0$$

$$\sigma_{yy} \sim \delta_{xy} \frac{h}{L} - \text{мало} \Rightarrow$$

$$\Rightarrow \sigma_{yy} \times \sigma_{zz} = 0$$

$$U = \int u dV$$

$$u = \frac{1}{2} (\sigma_{xx} \epsilon_{xx} + \sigma_{yy} \epsilon_{yy} + \sigma_{zz} \epsilon_{zz} + \tau_{xy} \gamma_{xy} + \tau_{xz} \gamma_{xz}) = \frac{1}{2} E y^2 (\partial_x^2 w)^2 + \frac{1}{2} G (x^2 + y^2) (\partial_x \varphi)^2$$

$$u = \frac{1}{2} E y^2 (\partial_x^2 w)^2 + \frac{1}{2} G r^2 (\partial_x \varphi)^2$$

A-mu-ee.

$$U = \frac{1}{2} \left(\int_0^L E (\partial_x^2 w)^2 \int_A y^2 dA + \int G (\partial_x \varphi)^2 \int r^2 dA \right)$$

$$I_z = \int_A y^2 dA \quad J = \int_A r^2 dA$$

$$U = \frac{1}{2} \left(\int_0^L E I_z (\partial_x^2 w)^2 dx + \int_0^L G J (\partial_x \varphi)^2 dx \right)$$

$$T = \frac{1}{2} \int_0^L [\rho S \dot{w}^2 - 2\rho S e \dot{w} \dot{\varphi} + \rho I_p \dot{\varphi}^2] dx$$

$$U = \frac{1}{2} \int_0^L [EI_z (\partial_x^2 w)^2 + GJ (\partial_x \varphi)^2] dx$$

$$\mathcal{L} = \frac{1}{2} \rho S (\dot{w})^2 - \rho S e \dot{w} \dot{\varphi} + \\ + \frac{1}{2} EI_z (\partial_x^2 w)^2 - \frac{1}{2} GJ (\varphi')^2$$

Уп-е лагранжа гул ному

1) $w(x, t)$

$$\frac{\partial \mathcal{L}}{\partial w} - \partial_t \left(\frac{\partial \mathcal{L}}{\partial \dot{w}} \right) - \partial_x \left(\frac{\partial \mathcal{L}}{\partial \partial_x w} \right) + \\ + \partial_x^2 \frac{\partial \mathcal{L}}{\partial \partial_x^2 w} = 0$$

2) $\varphi(x, t)$

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_t \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} - \partial_x \frac{\partial \mathcal{L}}{\partial \partial_x \varphi} = 0$$

$$\frac{\partial \mathcal{L}}{\partial w} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{w}} = \rho \delta \ddot{w} - \rho \delta c \dot{\varphi}$$

$$\partial_t \frac{\partial \mathcal{L}}{\partial \ddot{w}} = \rho \delta \ddot{w} - \rho \delta c \dot{\varphi}$$

$$\frac{\partial \mathcal{L}}{\partial \partial_x w} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \partial_x^2 w} = -EI_z \partial_x^2 w$$

$$\partial_x^2 \frac{\partial \mathcal{L}}{\partial (\partial_x^2 w)} = -EI_z \partial_x^4 w$$

$$\rho \delta \ddot{w} - \rho \delta c \dot{\varphi} + EI_z \partial_x^4 w = 0$$

$$\rho I \delta \ddot{\varphi} - \rho \delta c \ddot{w} - GJ \partial_x^2 \varphi = 0$$

Ищут решение стоячей волны

$$w = W(x) e^{i\omega t}$$

$$\varphi = \Phi(x) e^{i\omega t}$$

$$-\rho \delta \omega^2 W + \rho \delta c \omega^2 \Phi + EI_z \partial_x^4 W = 0 \quad (1)$$

$$-\rho I_p \omega^2 \Phi + \rho \delta c \omega^2 W - GJ \partial_x^2 \Phi = 0 \quad (2)$$

Гр-уи: $w(0) = w(L) = 0$
 $w'(0) = w'(L) = 0$
 $\varphi(0) = \varphi(L) = 0$

Разложение по соств. функциям

$$(\Phi = 0) \quad EI_z \partial_x^4 U_n = \rho \delta \omega_{u,n}^2 U_n \quad (3)$$

Совств. функции, соответствующие $U_n(x)$ для
 Грузовое ($W = 0$)

$$-GJ \Phi_m'' = \rho I_p \omega_{\varphi,m}^2 \Phi_m$$

Вблизи резонанса полагаем
 $\omega_{u,n} \approx \omega_{\varphi,m} \approx \omega$

$$W(x) = A U_n(x) \quad \text{(принимается одна мод)} \\
\Phi(x) = B \Phi_m(x)$$

Объяснение:

$$W(x) = A_1 U_1(x) + A_2 U_2(x) + \dots$$

Разложение по базису ед. ед. решений. В резонансе б.м. преобразуется

Уравнение (1) на U_n , интегрируя по dx

$$-\int \rho \omega^2 A \underbrace{\int U_n^2 dx}_{1(\text{нормировка})} + \int \rho e \omega^2 B \underbrace{\int U_n \Phi_m dx}_{= \langle U_n, \Phi_m \rangle} +$$

$$+ EI_z A \underbrace{\int U_n \partial_x^4 U_n dx}_{\lambda_n^4} = 0$$

$$\text{Из (3)} \quad EI_z \lambda_n^4 = \int \rho \omega_{u,n}^2$$

$$(\omega_{u,n}^2 - \omega^2) A + e \omega^2 \langle U_n, \Phi_m \rangle B = 0$$

Аналогично для (2), учитывая $\int I_p$

$$(\omega_{\Phi,m}^2 - \omega^2) B + \frac{\rho}{I_p} e \omega^2 \langle \Phi_m, U_n \rangle A = 0$$

$$\gamma_1 = e \langle U_n, \Phi_m \rangle$$

$$\gamma_2 = \frac{\rho}{I_p} e \langle \Phi_m, U_n \rangle$$

$$\omega_u = \omega_{u,n} \quad \omega_\Phi = \omega_{\Phi,m}$$

$$(\omega_u^2 - \omega^2)A + \gamma_1 \omega^2 B = 0$$

$$(\omega_\varphi - \omega^2)B + \gamma_2 \omega^2 A = 0$$

$$\vec{z} = \begin{pmatrix} \omega_u^2 - \omega^2 & \gamma_1 \omega^2 \\ \gamma_2 \omega^2 & \omega_\varphi^2 - \omega^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0$$

Система имеет решение $\exists \Leftrightarrow \det \vec{z} = 0$: $\gamma = \gamma_1 \gamma_2$

$$(\omega_u^2 - \omega^2)(\omega_\varphi^2 - \omega^2) - \gamma^2 \omega^4 = 0$$

$$(1 - \gamma^2) \omega^4 - (\omega_u^2 + \omega_\varphi^2) \omega^2 + \omega_u^2 \omega_\varphi^2 = 0$$

$$1) \gamma = 0: \omega^2 = \omega_u^2 \text{ или } \omega^2 = \omega_\varphi^2$$

Вдвиги резонанса $\omega_u \approx \varphi \approx \omega_0$ —
— одна корня одинаков.

$$2) \gamma \neq 0, \omega_u^2 = \omega_0^2 = \omega_\varphi^2$$

$$(1 - \gamma^2) \omega^4 - 2\omega_0^2 \omega^2 + \omega_0^4 = 0$$

$$\omega_1^2 = \frac{\omega_0^2}{1 - \gamma} \quad \omega_2^2 = \frac{\omega_0^2}{1 + \gamma}$$

$$\omega_{1,2}^2 = \omega_0^2 (1 \pm \gamma)$$

$$f_{12} = f_0 \left(1 \pm \frac{f}{2} \right)$$

$$f_1 = 4070 \text{ PHz}$$

$$f_2 = 4175 \text{ PHz}$$

$$f_0 = \frac{\omega_1 + \omega_2}{2} = 4123 \text{ PHz}$$

$$df_0 = \frac{\omega_2 - \omega_1}{2} = 53 \text{ PHz} \quad \Bigg| \Rightarrow$$

$$\Rightarrow f = 1,3 \cdot 10^{-2}, \quad \frac{f}{2} \approx 6,4 \cdot 10^{-3}$$

$$(\sigma_f)^2 = (6,4 \cdot 10^{-3} f)^2$$